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ORIGINAL ARTICLE

Tooth brushing using toothpaste containing theaflavins reduces the oral pathogenic bacteria in healthy adults

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Abstract

Theaflavins (TFs) are the main bioactive component in black tea. At present, little effort has been done to evaluate the influence of TFs when included in the toothpaste on the diversity of oral microbiota. In this study, eighty samples collected from the saliva and supragingival plaque of 20 healthy adults using toothpaste with the absence or presence of TFs for a period of 4 weeks were used for the oral microbiome analysis by 16S rRNA gene sequencing. Alpha and beta diversity analysis showed that tooth brushing using the toothpaste with TFs significantly increased the microbial abundance in the saliva samples, and altered the oral microbiota obtained from the saliva and supragingival plaque. The linear discriminant analysis revealed that the use of toothpaste with TFs significantly reduced the abundance of oral pathogens (e.g., *Prevotella*, *Selenomonas*, and *Atopobium*) while increased the abundance of oral-health associated bacteria (e.g., *Streptococcus* and *Rothia*). In addition, using toothpaste with TFs reduced the functional pathways abundance relevance to the extracellular polymeric substance (EPS) synthesis while enriched the functions in transporters, ABC transporters, two-component system, and amino acid metabolism. Collectively, our results provide evidence for the application of toothpaste containing TFs as a promising oral care product.

Keywords Theaflavins · Oral microbiota · Toothpaste · Black tea

Introduction

Oral microbiota, a predominant part of the human microbial library, plays an indispensable role in maintaining human health (Lamont et al. 2018; Rosier et al. 2018). Normally, the oral microbiota in healthy individuals live in symbiosis with the host, contributing to the healthy state by preventing the proliferation of pathogenic microorganisms (Rosier et al. 2018; Shang et al. 2020). Nevertheless, factors such as smoking, antibiotic treatment, sugar consumption, overexposure to fermentable carbohydrate, and oral hygiene are able to cause microbiota imbalance, which give the opportunity to the increased abundance of potentially pathogenic microorganisms, eventually resulting in oral diseases such as dental caries and periodontitis (Anderson et al. 2020; Rosier et al. 2018; Pitts et al. 2017; Al Bataineh et al. 2020). Dental caries is one common oral disease that threatens human health, which is mainly caused by the dominance of acidogenic microorganisms (e.g., *Streptococcus mutans*) (Lamont et al. 2018). To prevent the development of dental caries, oral care products containing antibacterial agents

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(e.g., stannous ions and chlorhexidine), including mouthwashes and toothpaste, have been widely applied (Anderson et al. 2020; Baena-Santillan et al. 2021). However, there are growing concerns about the widespread use of such broad-spectrum antimicrobial agents as they would contribute to the antimicrobial resistance (Philip et al. 2018; Wang et al. 2018). In addition, the overuse of antimicrobial agents will affect the community of health-associated bacteria and some agents have several side effects (Anderson et al. 2020).

In contrast to the traditional antibacterial agents, use of the extracts from natural plants has been a preferable strategy to prevent dental caries (Singla et al. 2018; Moraes et al. 2020). Tea prepared from the leaves of plant *Camellia sinensis* is one of the most important functional foods and is the most popular beverage after water throughout the world, which mainly includes three types on the basis of difference during the manufacturing process (green tea, oolong tea, and black tea) (Pinto 2013; Xing et al. 2019). The health benefits of tea consumption have been widely studied and documented, including the antioxidant, antimicrobial, anti-inflammatory, anti-allergic, and anti-carcinogenic activities (Tang et al. 2019). It is well-known that green tea consumption contributes to the maintenance of oral health and is effective in preventing the development of dental caries (Narotzki et al. 2012; Goenka et al. 2013; Nanri et al. 2019). A recent study revealed that the healthy benefits of green tea consumption are related to its impact on the oral microbiome, wherein a higher green tea intake increases the richness and diversity of oral microbiota and shifts the community composition (Peters et al. 2018). In addition, green tea consumption for two weeks changes the salivary microbiota and reduces the functional pathways abundance relevance to carcinogenesis (Yuan et al. 2018).

In comparison with green tea, black tea is more widely consumed. The main bioactive component in black tea are theaflavins (TFs). There is evidence showing that TFs possess antibacterial activity towards the oral pathogens such as *S. mutans* and *Porphyromonas gingivalis* (Wang et al. 2019; Lombardo Bedran et al. 2015). Thus, it suggests that TFs have beneficial effects on the oral microbiota related to the oral health. At present, the influence of TFs contained in oral care products on the oral microbiota remains elusive. For this reason, the current study aimed to assess the effect of tooth brushing using the toothpaste containing TFs on the oral microbiota in samples collected from the supragingival plaque and saliva after a tooth brushing period of four weeks in healthy adults. By high-throughput sequencing using 16S rRNA gene, we found that tooth brushing using toothpaste containing TFs significantly reduced the abundance of oral pathogenic bacteria while increased the abundance of oral-health-related bacteria. Our results shed light on the use of TFs in toothpaste as an oral care product in preventing the development of oral diseases.

Materials and methods

Bacteria and culture condition

Streptococcus mutans ATCC 25175 was purchased from the Guangdong Microbial Culture Collection Center (GDMCC, accession no. GIM1.369). *S. mutans* ATCC 25175 was grown in brain heart infusion (BHI) medium at 37 °C using an anaerobic incubator (Memmert, Germany). The ratio of O_2, CO_2, and N_2 in the incubator was set as 2:1:17.

Preparation of theaflavins

The theaflavins were synthesized by the enzymatic reaction, which was performed as previously described with some modifications (Subramanian et al. 1999; Sang et al. 2004). Briefly, the polyphenol oxidase mixture was prepared by adding 10 mL phosphate-citrate buffer (pH 4.8) to the 16 g fresh tea leaves (*Camellia sinensis* Longjing43) that were obtained from the Hangzhou Tea Experimental Plantation (Xihu, Hangzhou, China), followed by grinding at 0 °C. The mixture was stored at − 4 °C before use. The green tea polyphenol (with purity > 90%, 2 g) purchased from the Zhejiang Dongfang Tea Technology Co., Ltd. (Xihu, Hangzhou, China) was dissolved in a mixture of 100 mL acetone-pH 4.8 phosphate-citrate buffer (1:10, v/v), followed by adding the polyphenol oxidase mixture. Then, the enzyme reaction was performed by water bath at 30 °C, with stirring at 360 rpm and oxygen flux at 0.5 L/min. One hour later, the reaction mixture was centrifuged at 5000 rpm for 5 min (4 °C). The supernatant was extracted using the identical volume of ethyl acetate for three times, followed by concentrating on a rotary evaporator (IKA, Germany) at 30 °C. After concentration, the residue was purified using the Sephadex LH 20 column and eluted with acetone–water (2:3, v/v), followed by lyophilisation to obtain the crude theaflavins. The purity of the obtained theaflavins was determined by High-Performance Liquid Chromatography (HPLC) analysis (Waters, MA, USA).

Minimum inhibitory concentration determination

The theaflavins (with purity > 60%) were prepared as mentioned above. The minimum inhibitory concentration (MIC) of *S. mutans* ATCC 25175 to TFs was determined as previously reported (Xu et al. 2020), with some modifications. Briefly, a 24-h overnight culture of *S. mutans* ATCC 25175 was diluted by 1:100 into fresh BHI medium containing TFs with serially diluted concentrations. The mixtures were added to 96-well plates (200 µL in each well) and incubated at 37 °C anaerobically for 24 h. The optical density (OD_{600})

was determined by a microtiter plate reader (Victor™X3, PerkinElmer, Waltham, MA, USA). The lowest TFs concentration that inhibited visible bacterial growth was defined as the MIC (OD_{600} change < 0.05).

Study design and participants

A schematic flow chart of this study was shown in Fig. 1. Twenty volunteers aged between 18 and 23 (the mean age was 21) were recruited from Zhejiang Gongshang University. The inclusion criteria were: good health, minimum age 18, minimum number of teeth 25, with no clinical evidence of caries experience (DMFT = 0), no antibiotic therapy or professional cleaning within one month before the start of the study. The exclusion criteria were: pregnancy, nursing mothers, diabetics, denture wearers, smoking within the last 6 months, medical conditions and/or regular use of any medication which might affect the outcome of the study, and the obvious signs of untreated caries/significant periodontal disease. This study was reviewed and approved by the ethics committee of Zhejiang Gongshang University. The methods were carried out in accordance with the approved guidelines. All participants provided written informed consent. The 20 volunteers were randomly divided into 2 groups, with 5 males and 5 females in each group. The control group was required to use the toothpaste without TFs, while the experimental group was required to use the toothpaste with TFs. At the time of baseline (week 0) and post-4 weeks (week 4), the samples collected from the saliva and supragingival plaque were used for the subsequent analyses, respectively.

Sample collection

To investigate the influence of the use of toothpaste containing TFs on the oral microbiota, two toothpastes either with TFs or without TFs were applied. The components of the toothpaste were listed in Supplemental Table S1, which was produced by Hangzhou Jiaojie Oral Health Products Co., Ltd. (Xiacheng, Hangzhou, China). On the sampling time, the participants were asked to refrain from oral hygiene for 24 h and not to eat or drink for 2 h prior to sampling. For the collection of saliva samples, the participants were first asked to drink a glass of 150 mL sterilized normal saline to rinse the mouth for 10 s, followed by keeping lips shut for 3 min. Then, the saliva was expectorated directly into a specimen tube. The supragingival plaque sample of each participant was scraped from the buccal surface of two molars (#3 and #19), two incisors (#9 and #25), and two premolars (#12 and #28) by Gracey curette according to the American Dental Association (ADA) tooth numbering system. Samples were divided into eight groups PC0 ($n = 10$), PC4 ($n = 10$), PE0 ($n = 10$), PE4 ($n = 10$), SC0 ($n = 10$), SC4 ($n = 10$), SE0 ($n = 10$), SE4 ($n = 10$) (Supplemental Table S2). After collection, samples were immediately frozen and stored at − 80 °C before use.

DNA extraction and polymerase chain reaction

DNA was extracted using the E.Z.N.A.® Soil DNA Kit (D4015, Omega, Inc., USA) according to the manufacturer's instructions. The total DNA was eluted using 50 µL elution buffer and stored at − 80 °C before use. The polymerase chain reaction (PCR) analysis was performed by LC-Bio Technology Co., Ltd. (Jianggan, Hangzhou, China). The V3-V4 region of the 16S rRNA gene was amplified using the slightly modified version of primers 338F (5′-ACTCCTACG GGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGG TWTCTAAT-3′) (Fadrosh et al. 2014). The 5′ ends of the primers were tagged with specific barcodes per sample and the universal sequencing primers. The PCR amplification process was performed in a total volume of 25 µL reaction mixture containing 25 ng of the template DNA, 12.5 µL PCR Premix (M0536L, Yitao, Shanghai, China), 2.5 µL of each primer (10 µM), and nuclease-free water. The PCR conditions included an initial denaturation at 98 °C for 30 s; 35 cycles of denaturation at 98 °C for 10 s, annealing at 54 °C

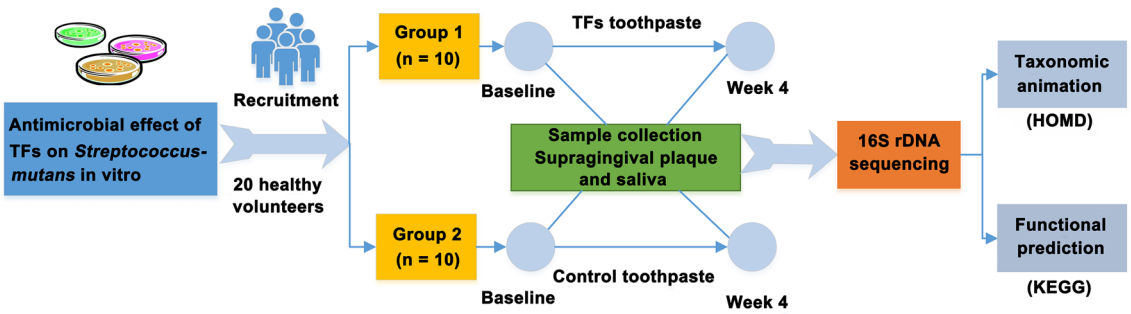


Fig. 1 Schematic flow chart of this study. Twenty healthy volunteers were divided into two groups randomly. The toothpastes with or without theaflavins were given to the members in each group. The saliva and plaque samples taken before (baseline) and after (4 weeks) the use of toothpaste were used for the subsequent analyses, including DNA extraction, 16S rDNA sequencing, and bioinformatics analysis

/52 °C for 30 s, and extension at 72 °C for 45 s; and then the final extension at 72 °C for 10 min. The PCR products were confirmed by 2% (w/v) agarose gel electrophoresis. Then, the PCR products were purified by AMPure XT beads (Beckman Coulter Genomics, Danvers, MA, USA) and quantified by Qubit (Invitrogen, Carlsbad, CA, USA). The amplicon pools were prepared for sequencing, while the size and quantity of the amplicon library were assessed by an Agilent 2100 Bioanalyzer (Agilent, Santa Clara, CA, USA) and the Library Quantification Kit for Illumina (Kapa Biosciences, Woburn, MA, USA), respectively.

High-throughput sequencing and data analysis

Samples were sequenced on an Illumina MiSeq platform (Illumina, San Diego, USA) according to the manufacturer's recommendations, which was carried out by the LC-Bio Technology Co., Ltd. (Jianggan, Hangzhou, China). Paired-end reads were assigned to samples based on their unique barcode and truncated by cutting off the barcodes and primer sequences. Paired-end reads were merged using FLASH (Fast Length Adjustment of Short reads, v. 1.2.8) (Magoc and Salzberg, 2011). Quality filtering on the raw tags were performed under specific filtering conditions to obtain the high-quality clean tags according to the fqtrim (v. 0.94). Chimera sequences were filtered by the Vsearch software (v. 2.3.4) (Rognes et al. 2016). Sequences with similarity $\geq 97\%$ were assigned to the same operational taxonomic unit (OTU) by Vsearch. Representative sequences were chosen for each OTU, and the taxonomic data were then assigned to each representative sequence using the Ribosomal Database Project (RDP) classifier (Cole et al. 2009). To study the phylogenetic relationship of different OTUs, multiple sequence alignments were conducted using the MAFFT software (v. 7.310) (Katoh et al. 2013). Alpha diversity was applied to analyze the complexity of species diversity for a sample through 4 indices, including Chao1, Shannon, Simpson, and the Observed species, which were calculated using the Quantitative Insights into Microbial Ecology (QIIME) software (v. 1.8.0) (Caporaso et al. 2010). Beta diversity analysis was used to evaluate the difference of samples in species complexity, which was performed by the principle coordinate analysis (PCoA). The Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) was applied to predict the functional content of the microbiota based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (Langille et al. 2013).

Statistical analysis was performed with R version 3.5.0. Differences among groups were analyzed using the analysis of variance (ANOVA) with Tukey's post-hoc test. Differences were considered statistically significant at $p < 0.05$. The linear discriminant analysis (LDA) effect size (LEfSe) analysis with the Kruskal–Wallis H-test was applied to detect the most discriminatory taxa among groups (Segata et al. 2011). Different features with an LDA score cut-off of 2.0 were identified.

Data availability

The raw reads had been deposited to the NCBI Sequence Read Archive (SRA) database (accession number: PRJNA684772).

Results

Determination of TFs concentration used in the toothpaste

To investigate the influence of TFs on the oral microbiota, we first determined the TFs concentration used in the toothpaste. Whereas that *S. mutans* is the representative species existing naturally in the oral cavity and responsible for the development of dental caries, we examined the antibacterial activity of TFs on the *S. mutans* ATCC 25175. We found that the MIC of *S. mutans* ATCC 25175 to theaflavins was 2.5 mg/mL. To evade the results bias that might be caused by the bactericidal activity of TFs, a sub-MIC of TFs (2.0 mg/mL) was used in the toothpaste.

Influence of TFs on the diversity of oral microbiota

To assess the changes in the oral microbiota after using the toothpaste containing TFs, a total of 80 samples (10 samples for each group) collected from the saliva and supragingival plaque were used for sequencing the V3-V4 region of the 16S rRNA gene. The raw data were filtered and sequences with a similarity greater than 97% were considered as one OTU. Overall, 1626, 442 raw reads and 4417 OTUs were obtained from all the samples, while the average number of OTUs in each group was 1648 (data not shown). A total of 16 phyla, 175 genera, and 445 species were detected from all the samples. The alpha diversity analysis using the Shannon index showed that the microbial abundance was significantly increased in SC4 and SE4 compared with that in the SC0 and SE0, suggesting that the use of toothpaste either with TFs or without TFs had a positive effect on the microbial abundance in saliva (Fig. 2a). The microbial abundance was significantly decreased in PC4 compared with that in the PC0, while the microbial abundance in PE4 was not significantly changed compared with that in the PE0. In addition, the use of toothpaste either containing TFs or not with a period of four weeks appeared to increase the number of bacterial species in the saliva and supragingival plaque samples (Fig. 2b), although the increase was not significant. The OTUs obtained from the groups SE0, SE4, SC0, SC4,

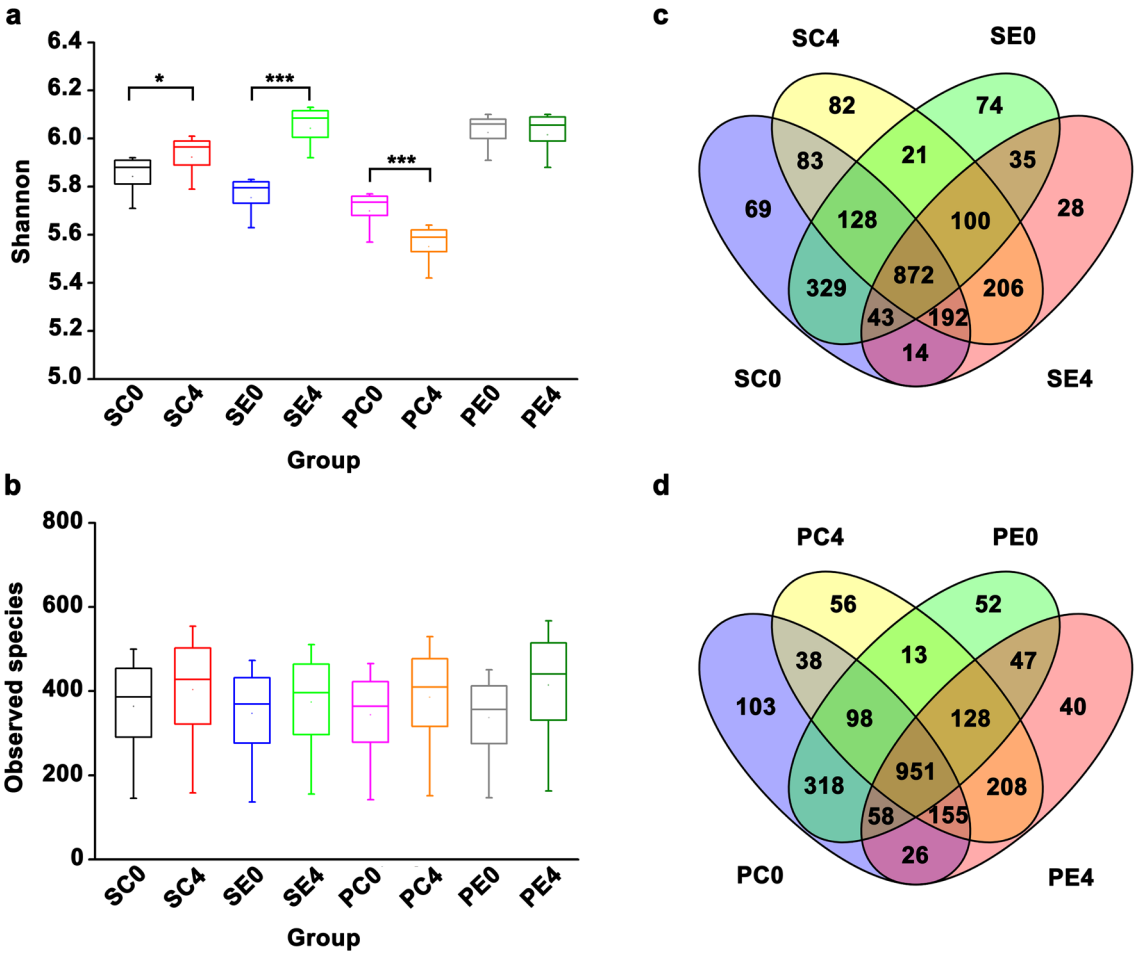


Fig. 2 Influence of tooth brushing using the toothpaste with or without TFs on the oral bacterial diversity. The alpha diversity was calculated using the Shannon index (**a**) and the Observed species index (**b**). **c** The Venn diagram among groups SC0, SC4, SE0, and SE4. **d** The Venn diagram among groups PC0, PC4, PE0, and PE4. Stars indicate significant differences ($p < 0.05$)

PC0, PC4, PE0, and PE4 were 1602, 1490, 1730, 1684, 1747, 1647, 1665, and 1613, respectively, among which the common OTUs obtained in the saliva and supragingival plaque samples were 872 and 952, respectively (Fig. 2c, d). Subsequently, the beta diversity analysis was performed to assess whether the overall bacteria taxa composition were different among different groups. The PCoA on unweighted unifrac distance showed that there was a significant separation between the groups (PC0, PE0, SC0, and SE0) and the groups (PC4, PE4, SC4, and SE4) ($p < 0.05$, analysis of similarities (ANOSIM), Fig. 3). Moreover, there was no significant separation between SE4 and SC4, as well as the PC4 and PE4, although individual variance was observed within each group.

Influence of TFs on the specific taxa abundance

We first analyzed the changes in the relative abundance of the most abundant taxa. The dominant phyla across all the

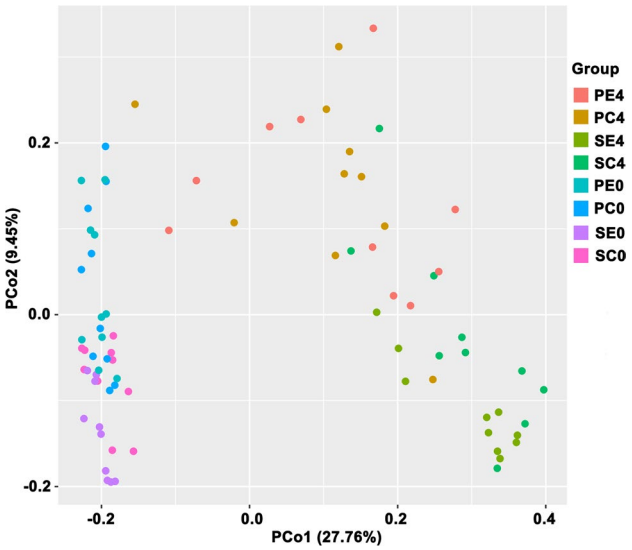


Fig. 3 Principal coordinate analysis (PCoA) based on unweighted unifrac distance

groups were *Firmicutes*, *Bacteroidetes*, *Proteobacteria*, *Actinobacteria*, *Fusobacteria*, SR1, *Spirochaetes*, *Saccharibacteria* (also known as *Candidate division* TM7), together comprising 97.0% of the total sequences (Fig. 4a). In the saliva samples, the relative abundance of *Bacteroidetes* was decreased in SC4 and SE4 compared with that in SC0 and SE0, while the abundance of *Proteobacteria* and *Actinobacteria* was increased in SC4 and SE4. The relative abundance of *SR1* and *Fusobacteria* was increased in SE4 compared with that in SE0. In the supragingival plaque samples, the relative abundance of *Bacteroidetes* and *Proteobacteria* was decreased in PC4 and PE4 compared with that in the PC0 and PE0, while the abundance of *Actinobacteria* and *Fusobacteria* was increased in PC4 and PE4. At the genus level, the main genera identified in this study were *Streptococcus*, *Prevotella*, *Haemophilus*, *Neisseria*, *Fusobacterium*, *Porphyromonas*, *Actinomyces*, *Alloprevotella*, *Rothia*, *Capnocytophaga*, *Gemella*, *SR1* genera, *Veillonella*, *Lautropia*, *Leptotrichia*, *Treponema,* and *Granulicatella*, together contributing > 80% to all the identified genera (Fig. 4b). The relative abundance of *Streptococcus*, *Rothia*, and *Lautropia* was increased in groups SC4, SE4, PC4, and PE4 compared with that in the groups SC0, SE0, PC0, and PE0, respectively, while the abundance of *Porphyromonas* and *Gemella* was decreased in groups SC4, SE4, PC4, and PE4. In addition, the use of toothpaste containing TFs reduced the relative abundance of *Alloprevotella*, *Capnocytophaga*, *Veillonella*, and *Granulicatella* while augmented the abundance of *Fusobacterium* and *Actinomyces* in the saliva and supragingival plaque.

We further determined the significantly changed genera upon the use of toothpaste containing TFs. The genera significantly enriched in the PE4 were *Fusobacterium*, *Corynebacterium*, *Actinomyces*, *Streptococcus*, *Rothia*, *Lautropia*, and *Lachnoanaerobaculum* compared with that in the PE0, while the relative abundance of *Haemophilus*,

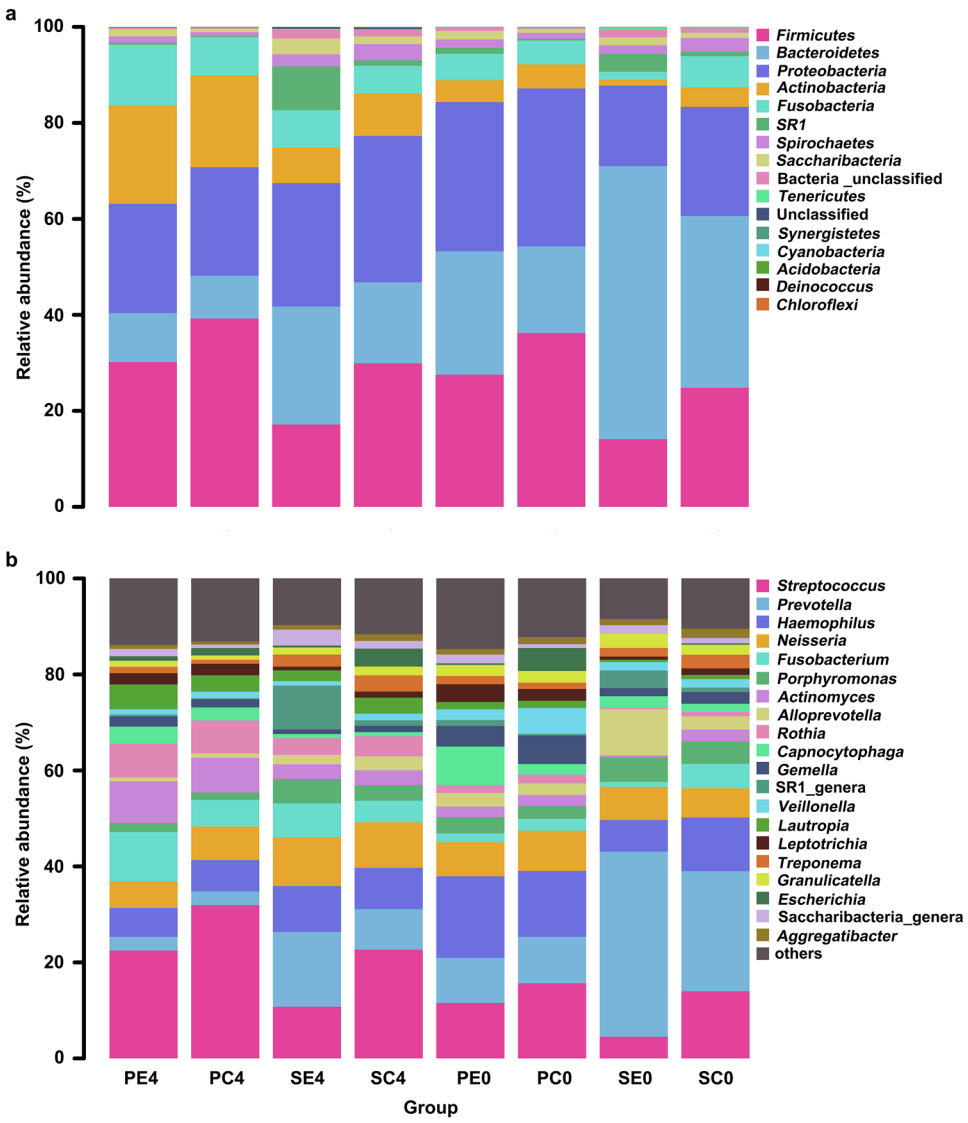


Fig. 4 The relative abundance of the identified bacteria in groups SC0, SE0, SC4, SE4, PC0, PE0, PC4, and PE4 at the phylum level (**a**) and genus level (**b**)

Granulicatella, *Gemella*, *Capnocytophaga*, *Alloprevotella*, *Veillonella*, *Selenomonas*, *Prevotella*, *Porphyromonas*, and *Leptotrichia* was significantly reduced in PE4 (Fig. 5). In the saliva samples, the use of toothpaste with TFs significantly decreased the content of *Capnocytophaga*, *Granulicatella*, *Atopobium*, *Alloprevotella*, *Veillonella*, *Selenomonas*, and *Prevotella* while increased the abundance of *Fusobacterium*, *Actinomyces*, *Streptococcus*, *Rothia*, and *Lautropia* etc. These findings were further verified by the LEfSe analysis, which were shown in the cladogram and the contributory discriminate taxa with LDA score > 3 was plotted for each group (Supplemental Fig. 1a, b; Supplemental Fig. 2a–c).

Functional prediction of the differentially abundant taxa

The PICRUSt (Langille et al. 2013) was applied to infer the functional content of the oral microbiota. We found an enrichment of the chaperones and folding catalysts, pores ion channels, membrane and intracellular structural molecules, lipopolysaccharide biosynthesis, and chromosome etc. in PE0 (Fig. 6, LDA score > 2.5), while an enrichment of the membrane and intracellular structural molecules, pyrimidine metabolism, chromosome, DNA repair and recombination proteins, starch and sucrose metabolism, and peptidases, was observed in the SE0. By comparison, the OTUs involved in the glycolysis, protein kinases, two component system, ATP-binding cassette (ABC) transporters, and transporters, and the OTUs included in the transporters, ABC transporters, bacterial motility proteins, two-component system, valine, leucine and isoleucine biosynthesis, porphyrin and chlorophyll metabolism, and arginine and proline metabolism etc. were increased in the PE4 and SE4 (Fig. 6), respectively.

Discussion

TFs modulate the oral microbiome but not the core microbiota

Daily tooth brushing using toothpaste is the most common way to maintain the oral health and a proper oral hygiene. However, the use of antibacterial toothpaste is not only able to inhibit the growth of oral pathogens but also produce unfavorable effects on the probiotic microorganisms (Shang et al. 2020). Therefore, the search for a compound that is effective in preventing the proliferation of oral pathogens while has negligible influence on the growth of commensal microorganisms is in demand. TFs are the predominant bioactive component of the black tea that is widely consumed around the world (Tang et al. 2019; Pinto 2013). In the past years, the antibacterial activity of TFs on the oral pathogenic bacteria that can cause oral disease, such as *S. mutans* and *P. gingivalis*, has been investigated by in vitro test (Lombardo Bedran et al. 2015; Wang et al. 2019). However, whether TFs could be added to the toothpaste as an oral care product and its influence on the oral-health associated bacteria remain elusive. In this study, our results suggest that TFs when included in the daily used toothpaste contribute to oral health. First, in consistent with the previous

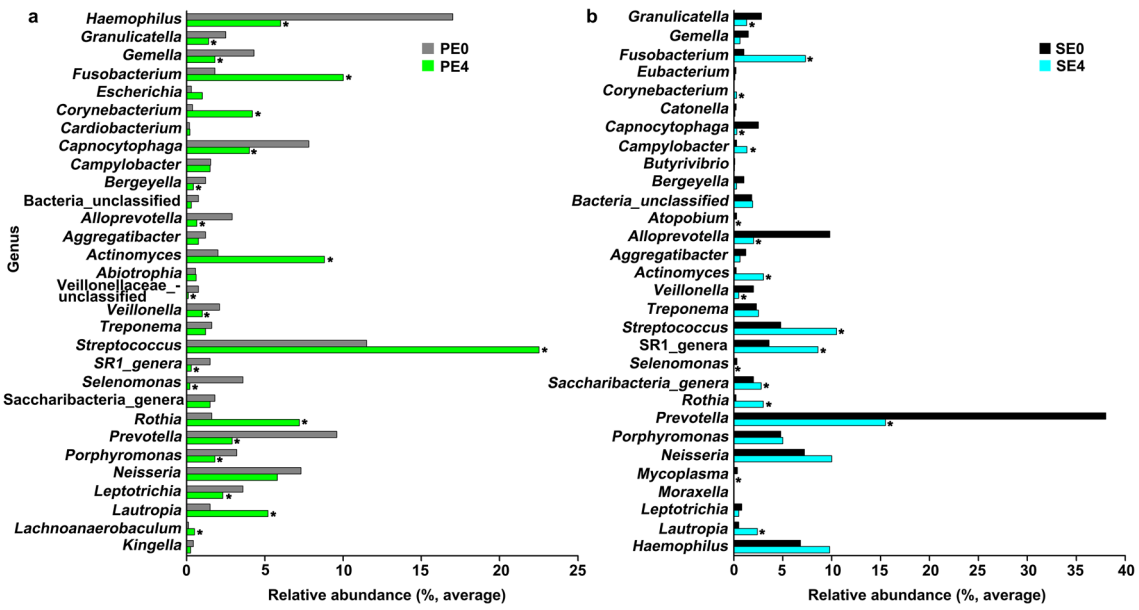


Fig. 5 The significantly changed bacteria upon the use of toothpaste containing TFs at the genus level. **a** Saliva samples. **b** Supragingival plaque samples

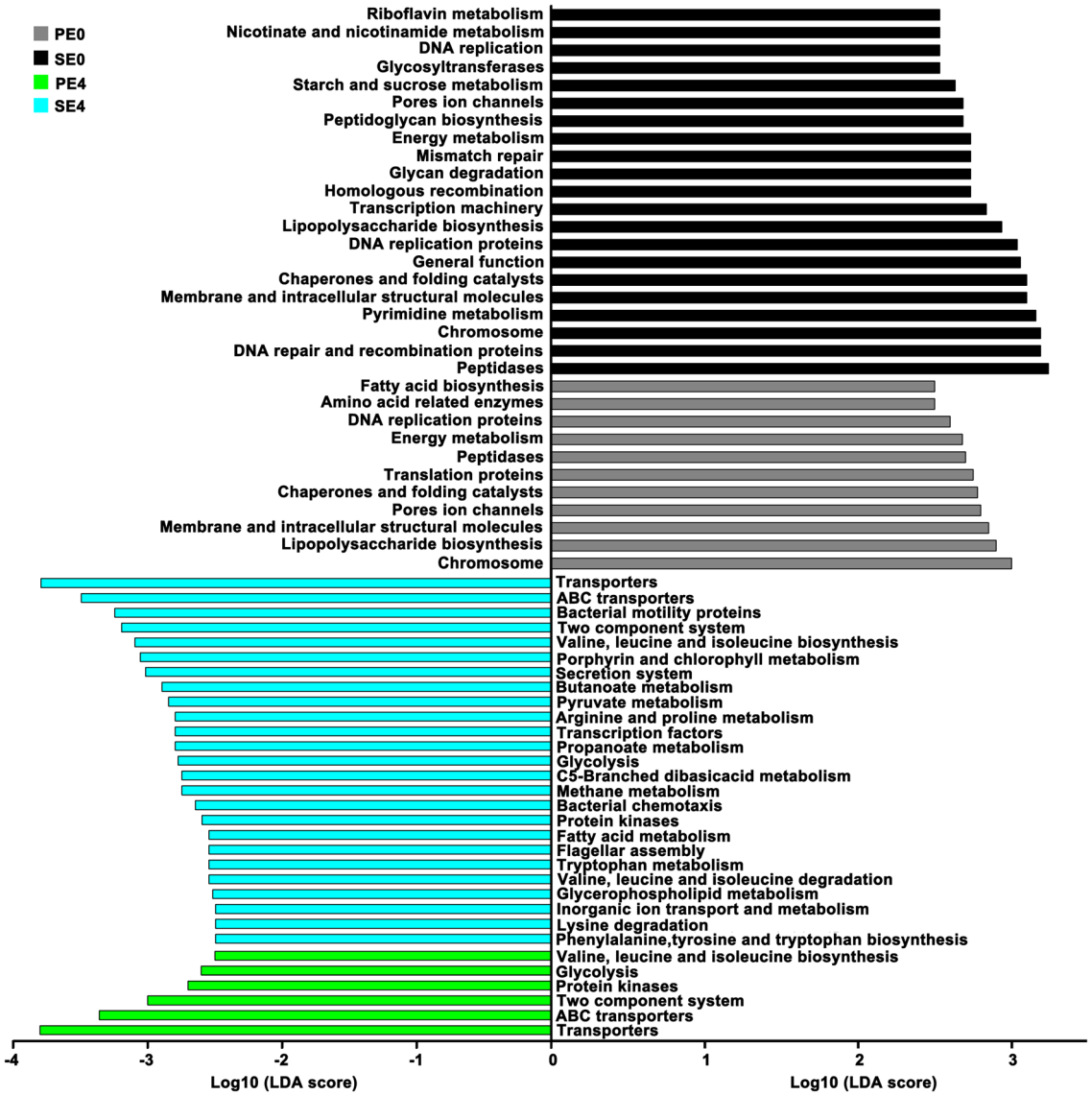


Fig. 6 Functional prediction of the oral microbiota obtained from the groups PE0, SE0, PE4 and SE4 using PICRUSt. Pathway abundances were predicted from the 16S rRNA gene sequencing data. Functional categories with a LDA score greater than 2.5 were shown

findings that TFs are effective in inhibiting the growth of *S. mutans* in vitro (Wang et al. 2019), we found that the MIC of *S. mutans* ATCC 25175 to TFs was 2.5 mg/mL. Second, tooth brushing using the toothpaste containing TFs with sub-MIC did not reduce the microbial abundance of the saliva samples. Instead, the microbial abundance was significantly increased after the tooth brushing with a period of four weeks. In addition, most of the dominant phyla and OTUs were common in either SE0 and SE4 or PE0 and PE4, indicating a core microbiota was not affected by the use of toothpaste with TFs. This further suggests that TFs are likely to modulate the already existed microbial flora rather than to build a completely novel composition of the oral microbiota. Third, tooth brushing using toothpaste containing TFs produced a negligible effect on the abundance of oral health-associated bacteria while reduced the diversity of oral pathogenic bacteria.

TFs increase the abundance of oral-health associated bacteria while reduce the abundance of oral pathogens

In this study, the 5 most abundant phyla identified in all the samples were *Firmicutes*, *Bacteroidetes*, *Proteobacteria*, *Actinobacteria*, and *Fusobacteria*, which was in line with the previous oral microbiome studies wherein these five phyla are the most common bacteria in people with age from early child to elderly adults (Xu et al. 2018; Jiang et al. 2018;

Xiao et al. 2016), suggesting the steady microbial composition in people with different lifestyles. There are evidences showing that species belonging to the genera *Veillonella*, *Prevotella*, *Selenomonas*, *Atopobium*, *Granulicatella*, *Leptotrichia*, and *Alloprevotella* are closely associated with the development of dental caries (He et al. 2018; Hurley et al. 2019; Xiao et al. 2016). It is worth noting that tooth brushing using the toothpaste containing TFs significantly reduced the abundance of the above-mentioned genera both in the saliva and the supragingival plaque after a period of four weeks (Fig. 5). *S. mutans* is one notorious microorganism that has long been recognized as the pathogen responsible for dental caries (Lamont et al. 2018). However, this species was not found in nearly all of the samples. This bias may result from its low abundance in the saliva and supragingival plaque since the participants we recruited in this study were all in a caries-free state.

In addition to the oral pathogens, we found that the abundance of genera *Fusobacterium*, *Corynebacterium*, *Actinomyces*, *Streptococcus*, *Rothia*, and *Lautropia* was significantly increased in SE4 and PE4 compared with that in SE0 and PE0, respectively. Notably, the bacterial genera *Fusobacterium*, *Corynebacterium*, *Actinomyces*, and *Streptococcus* have previously been shown as the dominant genera in the tooth and gum tissue with a healthy state (Liu et al. 2012). Moreover, a previous study found that the abundance of *Corynebacterium* bacteria was decreased significantly in the supragingival plaques of dental caries (Xiao et al. 2016). For the *Streptococcus*, there is no question that *Streptococcus* is one of the most prevalent and common genus found in the oral cavity, although some *Streptococcus* species are the driving force of dental caries (Lamont et al. 2018). In a recent study, *Rothia* is suggested to be related to the dental health since its abundance was decreased significantly with the development of dental caries (Xu et al. 2018). Collectively, our results suggest that tooth brushing using the toothpaste containing TFs is able to reduce the abundance of oral pathogens while increase the abundance of oral-health-associated bacteria.

Tooth brushing using toothpaste containing TFs alters oral microbiota functions

Our results suggest that the bacterial functions are changed upon the use of toothpaste with TFs (Fig. 6). It is known that the development of dental caries is driven by diet–microbiota interactions, wherein an overexposure to fermentable carbohydrates is prone to result in the increased synthesis of extracellular polymeric substance (EPS) (Lamont et al. 2018). EPS including exopolysaccharides such as glucans, extracellular DNA (eDNA), and glycoproteins are critical for the promotion of microbial shifts towards acidogenic and aciduric organisms, eventually leading to the caries development (Bowen et al. 2018; Takahashi and Nyvad 2011). Notably, the observed functional enrichments such as DNA synthesis and replication, sucrose and starch metabolism, glycan degradation, lipopolysaccharide biosynthesis, and glycosyltransferases that are associated with the EPS production in PE0 and SE0 were not shown in PE4 and SE4. This indicates that TFs may aid in preventing caries development by interfering with the EPS production.

It is worth noting that after the use of toothpaste with TFs for 4 weeks, the main bacterial functions were enriched in transporters, ABC transporters, and two-component system (TCS) in the saliva and supragingival plaque samples. Membrane channels, such as ABC transporters and the major facilitator superfamily (MFS) transporters, are critical for up-taking nutrients and extruding deleterious compounds in bacteria (Wang et al. 2020; Theodoulou and Kerr 2015). Thus, the functional enrichment in transporters suggests an increased level in the metabolic activity of the beneficial bacteria. In *S. mutans*, ABC transporter proteins have been shown to be responsible for the survival of *S. mutans* under stressful conditions (Biswas and Biswas 2011; Nagayama et al. 2014), which further indicates that TFs have an influence on the transport system of the beneficial bacteria. TCS is one critical signal pathway responsible for sensing the environmental cues and regulating adaptive responses (Stein et al. 2020). There is evidence showing that TCS is important for the *S. mutans* to defend against human- or bacteria-derived antibacterial peptides (Kawada-Matsuo and Komatsuzawa 2017). Therefore, functional enrichment in the category of TCS suggests that the oral microbiota regulate their gene expression profile in response to TFs treatment. In addition, amino acid metabolism was found to be enriched in the SE4 and PE4. Amino acid metabolism such as arginine metabolism is well-known to have a contribution to the neutralization of oral acidic environment caused by the acidogenic bacteria (He et al. 2016; Chakraborty and Burne 2017; Huang et al. 2015), wherein arginine can be catabolized by the arginine deiminase system (ADS) to release ammonia to raise the pH of the oral environment. This process is beneficial for the growth of health-associated bacteria (Jakubovics et al. 2015). From this point of view, the reduced abundance of oral pathogens upon the use of toothpaste with TFs may be partly achieved by increasing the pH of the oral cavity.

In conclusion, our findings reveal that tooth brushing using toothpaste containing TFs is able to produce a beneficial shift in the taxonomic abundance of oral microbiota, with a reduction in the pathogenic bacteria and an increase in the oral-health related bacteria. Moreover, the use of toothpaste containing TFs has little effect on the core oral microbiota. Thus, TFs when included in the toothpaste are suggested to be helpful for the improvement and protection of oral health. Nevertheless, we need to consider three issues that were not investigated in the current study. First, the

volunteers we recruited are all in a healthy state, thus if the use of toothpaste containing TFs is capable of ameliorating the symptom of oral diseases already existed needs further analysis. Second, the results obtained in this study are from short-term use of the toothpaste with TFs, thus the influence of long-term use of this toothpaste on the oral microbiota needs future assessment. Third, for the use of TFs concentration in the toothpaste, we only tested 2.0 mg/mL based on the antibacterial activity of TFs on the *S. mutans* ATCC 25175. Thus, the influence of TFs with different concentrations on the oral microbiota needs future investigation.

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Author contributions JK, GZ, and CD conducted the lab work and data analysis. KX and JK interpreted the data and drafted the manuscript. XY and XZ contributed partly to the lab work. XL, KX, and YL conceived of the study, performed data review, and contributed to the manuscript writing. All authors have read and approved the final manuscript.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This study was reviewed and approved by the ethics committee of Zhejiang Gongshang University. The methods were carried out in accordance with the approved guidelines. All participants provided written informed consent.

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